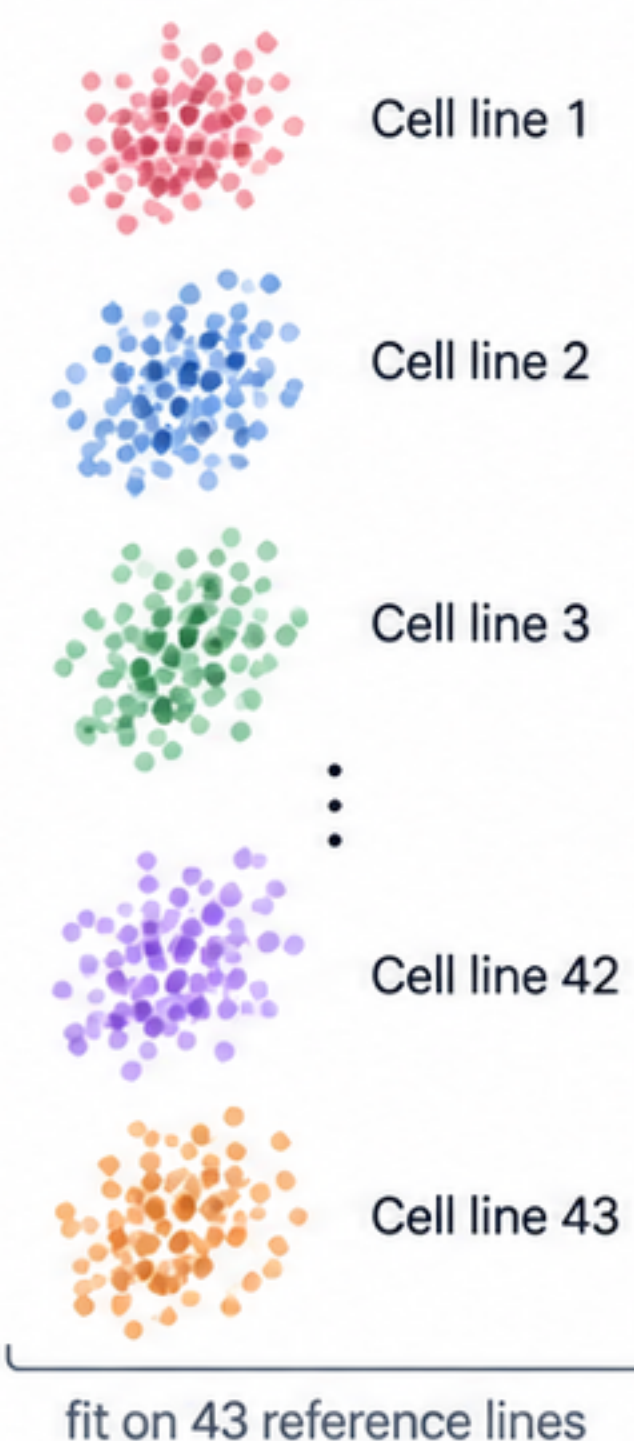
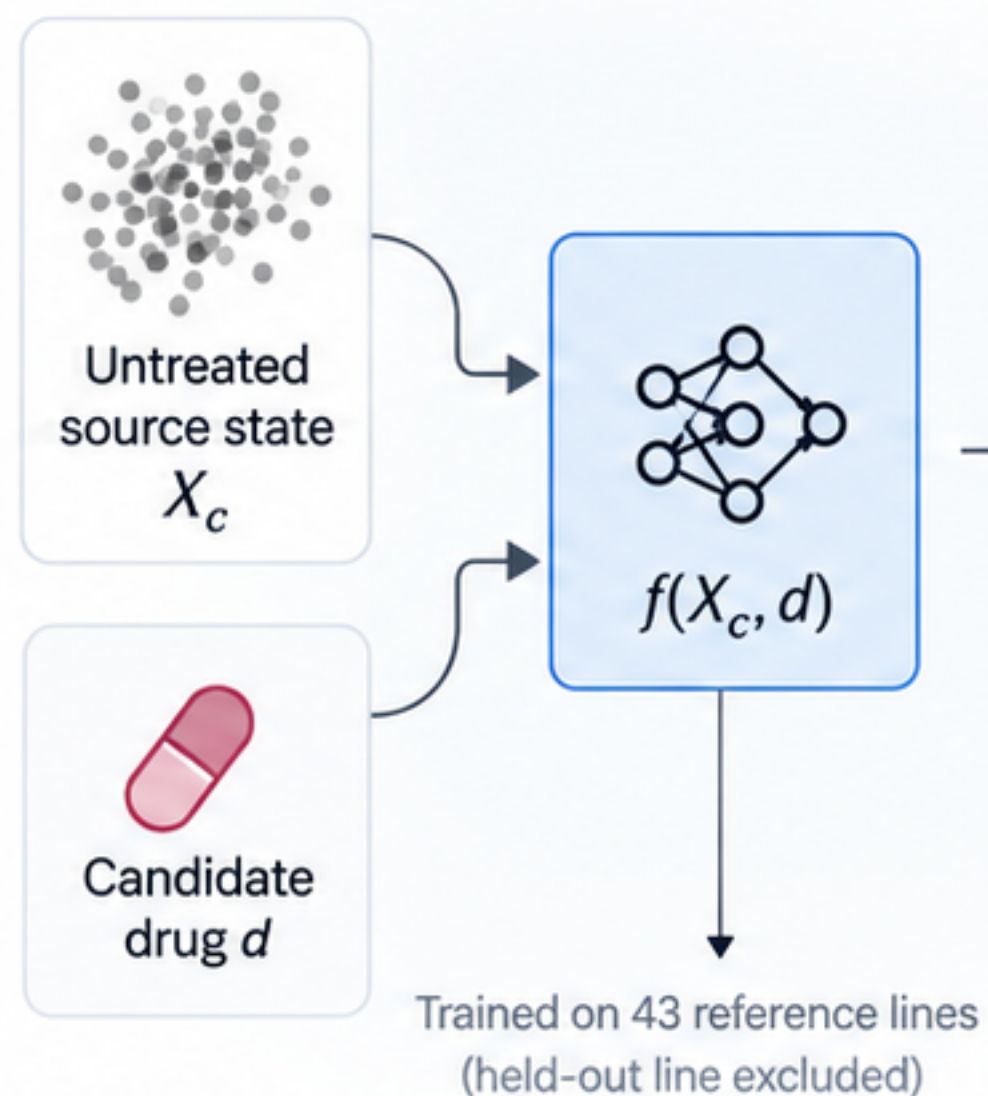


a**Training contexts**

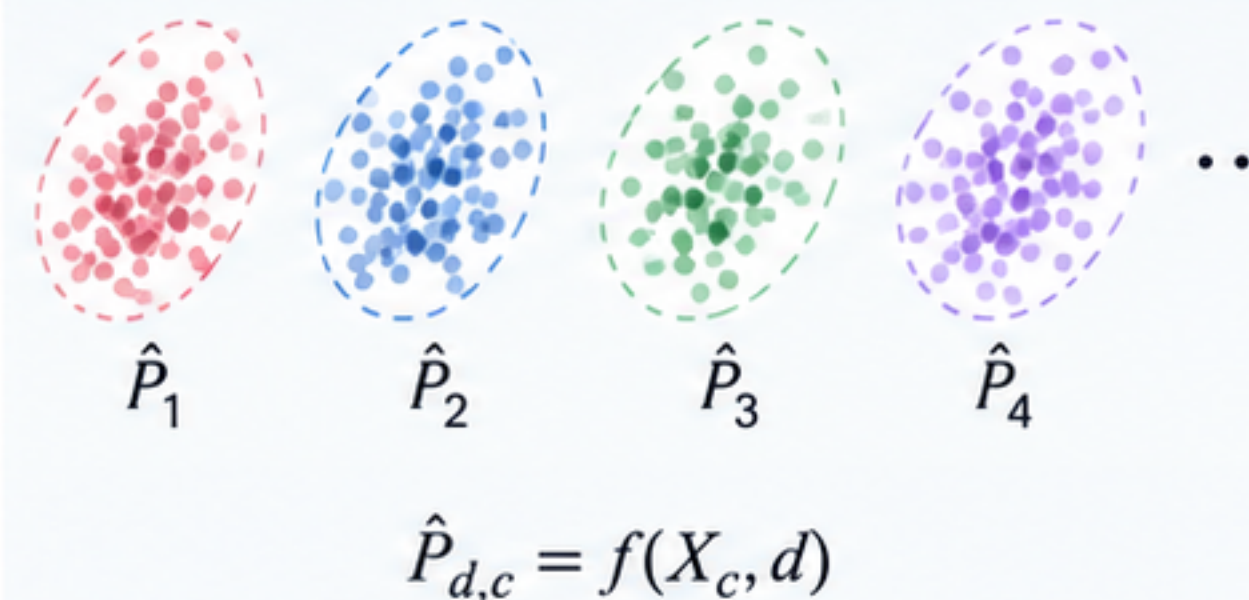
43 reference cell lines
observed drug responses

**Forward response predictor**
 $f(\text{source state, drug})$
**Held-out cell line**

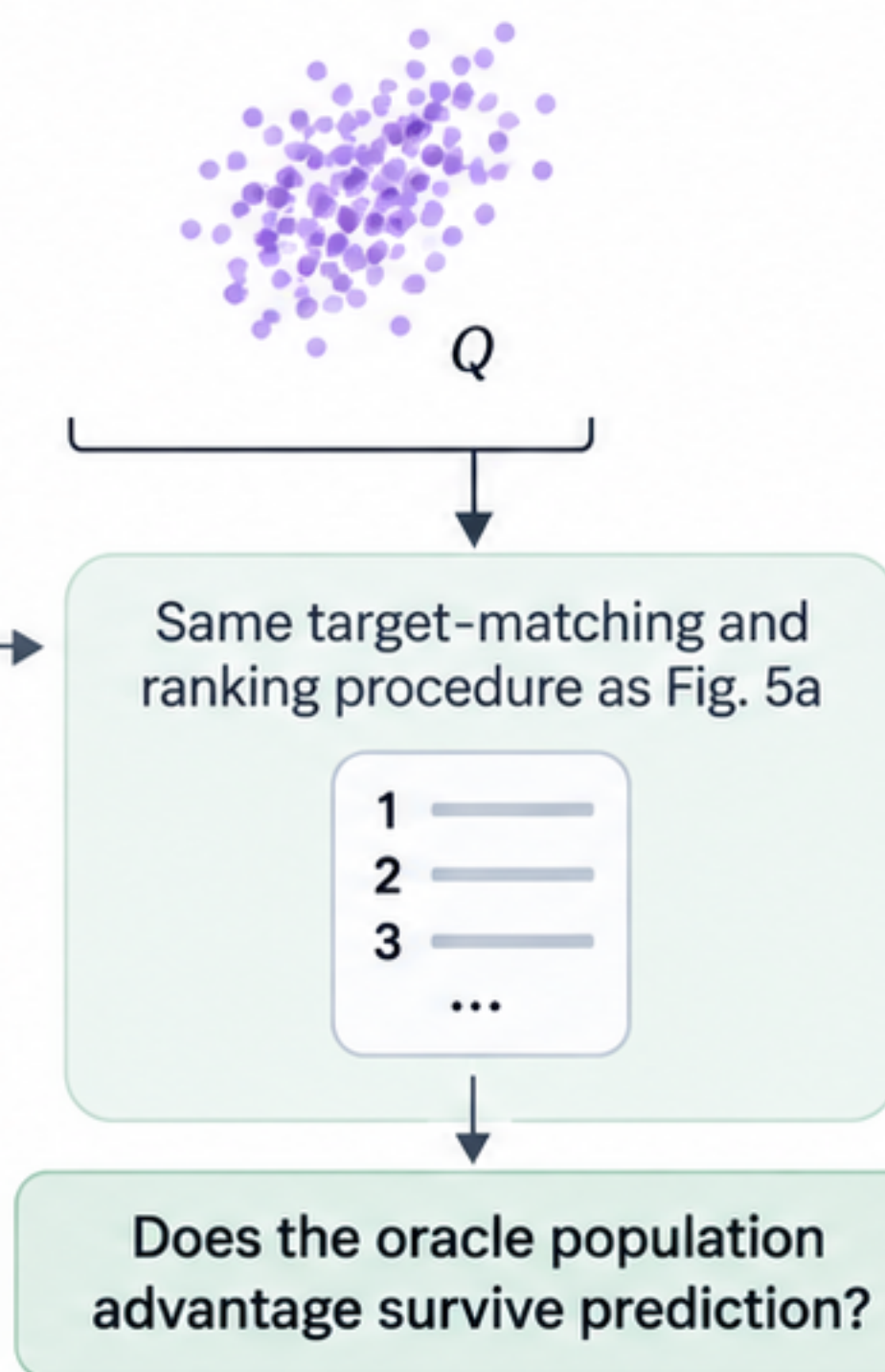
92 fixed candidate drugs

**Predicted candidate responses**

predicted in the held-out line, not observed

**Target population Q**

Wanted state

**Prediction-constrained intervention-identification task in a held-out cell line**

Candidate outcomes are predicted rather than observed; the target population is given only to the retrieval scorer.